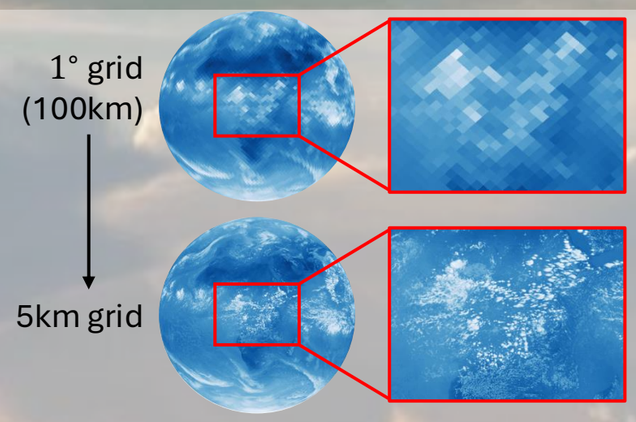
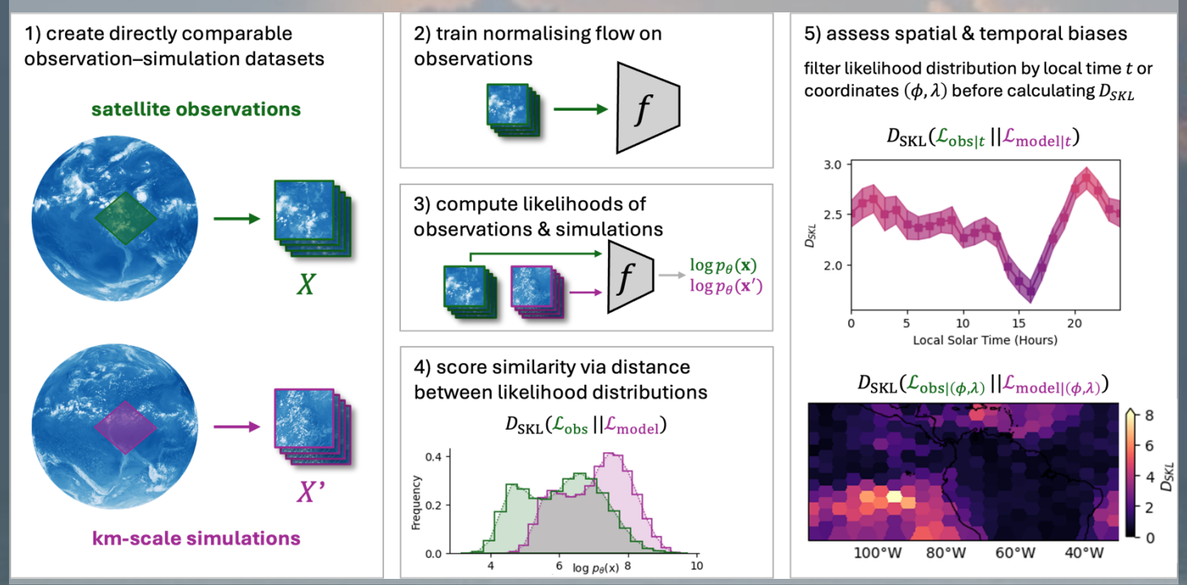
Evaluating the representation of deep convective clouds in global km-scale climate models

Lilli Freischem, Tim Reichelt, Philip Stier & Hannah Christensen (University of Oxford)

Motivation:
we need new metrics to evaluate km-scale models based on high-resolution cloud features

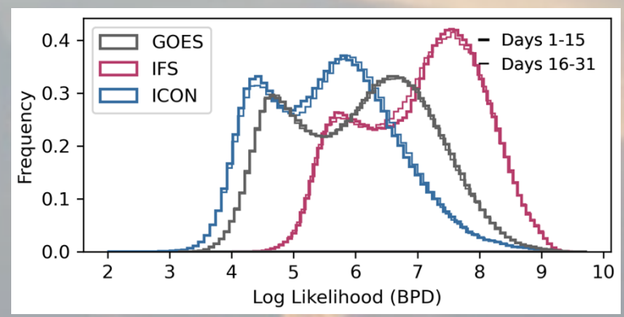


A generative likelihood framework for evaluating km-scale climate models

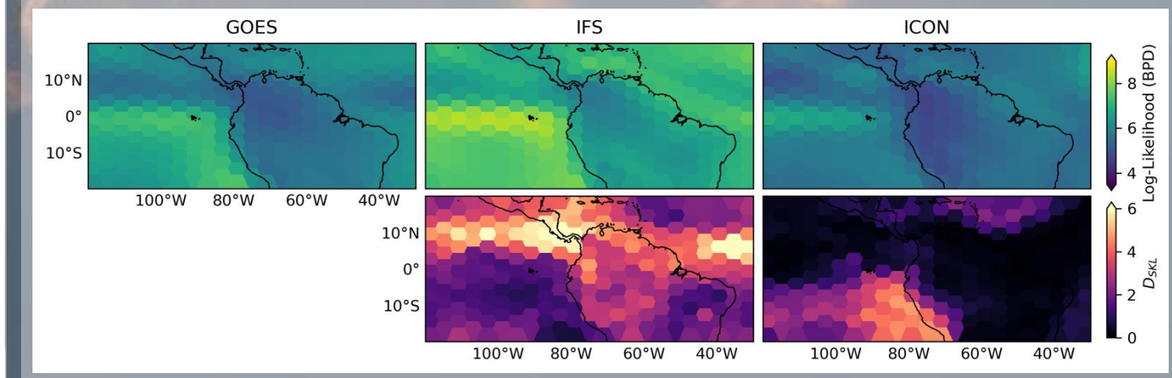


Biases in outgoing longwave radiation of two km-scale climate models IFS and ICON, compared to GOES-16 geostationary satellite observations.

Histograms of log-likelihoods (bits/dim) under the normalising flow trained on GOES observations.



Spatial patterns of biases in IFS and ICON.



Ongoing work: intercomparison of deep convective clouds across kilometre-scale models.

(Research) Interests: machine learning for climate modelling & earth observation, (deep) convective organisation.
In my spare time, I can be found outdoors, usually in a rowing boat.